

When study design predicts disease: an identifiability limit for patient-level single-cell classifiers

Hongyu Ying¹, Dandan Yun¹ and Dan Liu^{1,*}

¹ Department of Rheumatology and Immunology, Shanghai Pudong Hospital, Fudan University Pudong Medical Center, Shanghai 201399, China * Correspondence: Dan Liu; danliu600@126.com

Abstract

Single-cell RNA sequencing resolves disease-associated immune states at cellular resolution, but clinical classification concerns patients. Patient-level models compress thousands of cells into one donor representation and can report cross-validated discrimination above 0.95. Such performance is ambiguous when recruitment, processing or sequencing design also predicts the disease label.

We define a design-label estimability scale and test it in three systemic lupus erythematosus cohorts. In an additive model, the fraction of label information remaining after design adjustment is $1-R^2(Y \sim D)$. Across 11,200 simulations, increasing design-label association depleted this information, destabilised biological-effect recovery and created internal-to-external gaps. A low-entanglement positive control passed all five proposed validity checks.

The empirical cohorts occupied the failure regime. In GSE135779, recorded design, sequencing and demographic variables predicted disease at AUC 0.952 and exceeded 1,000 complete-pipeline label permutations ($p=0.000999$). Nonlinear design-only models also discriminated disease (AUC 0.815-0.824). Yet batch provided an internal negative control: representations identified batch at AUC up to 0.993, while batch alone predicted disease at 0.499 and batch residualisation changed disease AUC by at most 0.009. In GSE174188, design restriction preserved most discrimination whereas linear residualisation removed 0.230-0.268 AUC. Random-forest residualisation in GSE135779 also attenuated all representations, while batch location adjustment and overlap weighting largely preserved them. Strict source-only transfer was weaker than internal validation; pseudobulk remained strongest in the 261-donor target and learned pooling did not repair the gap.

We convert these results into five executable checks spanning estimability, full-pipeline nulls, design exposure, overlap restriction and external transfer. High internal AUC establishes separability; attribution requires design overlap, and transport requires an independent acquisition process.

Keywords: patient-level classification; single-cell RNA sequencing; identifiability; study design; batch confounding; residualisation; external validation; Geneformer; pseudobulk; systemic lupus erythematosus

Author Summary

Researchers use single-cell sequencing to build computer models that predict whether a patient has a disease. These models often appear highly accurate. We asked a simple question: are they detecting the disease itself, or are they detecting how the samples happened to be collected and processed?

We studied two lupus datasets. In both, information about study logistics alone, including when a sample was collected, which processing group it belonged to and its sequencing quality, predicted disease without using any biological measurement. When study design and disease are closely linked, an analysis of that dataset alone cannot separate the two explanations.

We found that two standard remedies disagree sharply. Statistically subtracting design effects removed far more apparent accuracy than comparing patients who were processed under matched conditions. We verified that this disagreement remained with a nonlinear adjustment method. A separate internal control showed why simply detecting processing information in a model does not prove that the model used it to predict disease.

We propose five checks that researchers can run before claiming that a patient-level model has detected biology. Our code, data, software environments and automated checks accompany the manuscript so others can apply the protocol.

1. Introduction

Systemic lupus erythematosus is a heterogeneous autoimmune disease in which blood interferon activity and immune-cell states vary substantially between patients. Single-cell RNA sequencing can resolve those states, but diagnosis concerns a patient while the measurements are made in individual cells. Patient-level modelling must therefore compress a variable-sized cell population into one representation without confusing biological signal with the way the cohort was assembled.

Several solutions now compete for this role. Pseudobulk aggregates expression, CloudPred and singleDeep learn phenotype-oriented summaries, and recent methods such as PaSCient, scPhase and FloREN learn multicellular or graph-based patient representations [6-11]. Frozen single-cell foundation models offer a particularly attractive shortcut: encode each cell once, average the embeddings, and train a small donor classifier. This strategy is computationally convenient, but high within-cohort accuracy does not establish that the representation will survive a change of hospital, population, acquisition period or sequencing workflow.

The hidden difficulty is that many public disease cohorts are not random samples from one acquisition process. Cases and controls may come from different recruitment sources, years, multiplexing pools or processing waves. Batch-label confounding has long been known to bias cross-validation [16], and a completely confounded single-cell design is statistically non-identifiable [17]. This is not merely a preprocessing problem. If disease and design are collinear, the observed data cannot distinguish a biological disease effect from a technical design effect without additional overlap or external information. Regressing a representation on technical covariates does not restore that missing contrast: it removes the disease-aligned component whether that component is technical, biological or both [18,19].

This problem has become more urgent, not less. Broad benchmarks show that frozen single-cell foundation-model embeddings can retain batch structure and can lose to simple expression baselines [13,14]. Patient-level methods already report high internal performance in SLE, including external validation between GSE174188 and GSE135779 [7]. FloREN now evaluates both cohorts with supervised graph representations [11], while pretraining-contamination work argues that benchmark validity must accompany model rank [15]. What remains missing is a patient-level framework that connects the estimability limit to executable diagnostics, then tests whether design predictability, residualisation, design restriction and external transfer tell the same story on the same donors. They do not.

Predictive-modelling research has already shown that confound adjustment must be contained within training folds, that featurewise confound regression can leak or reveal information to nonlinear learners, and that residual confounding can persist after apparently successful mitigation [27-29]. Permutation-based confounding tests and classifier-dependent indices provide complementary diagnostics [25,30]. These results motivate caution, but they do not quantify how much disease-label contrast remains estimable from a patient-level single-cell cohort, nor compare adjustment with observed-overlap restriction and source-only transfer. A recent unsupervised single-cell benchmark further established cell-type composition as a strong patient-stratification baseline [26], making composition an essential rather than optional control.

Here we define a design-adjusted label-information measure, construct observational equivalence at perfect design-label collinearity and map the resulting failure regime in simulation. We then recover donor design metadata for GSE135779, combine it with cell-distributed processing metadata in GSE174188, and test frozen Geneformer plus two expression pseudobulk representations. Design-only prediction, complete-pipeline permutation, representation-to-design prediction, fold-contained residualisation, overlap restriction, matched donor controls, cross-batch transfer, cross-cohort transfer and learned pooling are evaluated under one donor-matched framework. This moves beyond another representation leaderboard: it defines when a patient-level disease signal can be separated, when it can be attributed and when it can be expected to transport.

2. Results

2.1. Study design predicts disease in two widely reused SLE cohorts

GSE174188 contains 261 analysed CD4-positive donors: 162 SLE cases and 99 controls [2]. The distributed single-cell object records four processing waves. In the CD4 subset, their control/case counts were 32/0, 17/97, 7/19 and 43/46 (Figure 1A). Wave 1 contains no cases, and wave 2 is dominated by cases. A logistic model using only donor processing-wave fractions achieved AUC 0.922 (2.5th-97.5th percentile over 20 repeated five-fold donor analyses: 0.916-0.928). Sequencing/QC variables reached 0.781, and the complete measured design block reached 0.953 (Table 1; Figure 1D).

For GSE135779, we restored batch, collection year, demographics, clinical fields and sequencing QC for all 44 childhood-cohort donors from the original Supplementary Tables 1b and 1c and mapped them to the analysis identifiers through GEO [1]. No donor or selected field was missing. The six sequencing batches contained case/control counts of 8/0, 3/1, 6/2, 6/2, 6/2 and 4/4 (Figure 1B). Collection years 2014 and 2016 contained only cases, whereas 2018 was balanced at 8/8 (Figure 1C). Collection year alone predicted disease at AUC 0.721, demographics at 0.715 and sequencing/QC at 0.707. Their combination with batch reached AUC 0.952 (0.893-0.992).

These two cohorts encode design-label entanglement differently. GSE174188 has one dominant processing coordinate that is itself a strong disease classifier. GSE135779 distributes the label channel across year, sequencing and demographic variables. A batch table alone would expose the first design but miss the second.

Table 1. Cohort design structure and design-only discrimination.

Cohort	Donors (case/control)	Recorded design	Complete-design AUC	Most informative single block
GSE135779	44 (33/11)	6 batches; 4 collection years; sequencing QC; demographics	0.952 [0.893-0.992]	collection year: 0.721
GSE174188 CD4	261 (162/99)	4 processing-wave fractions; sequencing QC; age; sex	0.953 [0.945-0.961]	processing wave: 0.922
GSE285773 CD4	26 (16/10)	donor labels and cell profiles; no recoverable donor processing batch	not estimated	not estimated

2.2. A quantitative estimability scale for design-label association

Let Y denote the centred donor disease label, D the recorded design matrix including an intercept, and M_D the projection onto the orthogonal complement of the column space of D . The fraction of label information remaining after linear design adjustment is

$$I_D = (Y^T M_D Y) / (Y^T M_1 Y) = 1 - R^2(Y \sim D).$$

This scale is one when design is unrelated to the label and zero when design perfectly spans it. In the additive model, relative variance for estimating a disease coefficient from the remaining contrast increases as $1/I_D$. At perfect collinearity, a biological-only mechanism $X = \beta Y + \epsilon$ and a technical-only mechanism $X = \delta D + \epsilon$ can produce exactly the same observed (X, D, Y) distribution. The construction is a direct consequence of collinearity; its value here is to turn that familiar fact into a cohort-level scale and a testable patient-representation protocol.

We mapped this boundary in 11,200 donor-level simulations crossing four biological effects, four technical effects and seven levels of design-label association (Figure 2). Mean I_D fell monotonically from 0.996 at zero association to 0.189 at $\rho = 0.90$, 0.039 at $\rho = 0.98$ and zero at perfect collinearity. Median variance inflation reached 5.27 at

$\rho=0.90$ and 30.18 at $\rho=0.98$. With a purely technical signal ($\beta=0, \delta=2, \rho=0.90$), internal unadjusted AUC was 0.936, while residualised, overlap-restricted and independent-target AUCs were 0.499, 0.501 and 0.499. With mixed biological and technical signal ($\beta=1, \delta=1, \rho=0.90$), internal AUC was 0.947, residualisation reduced it to 0.573, overlap restriction retained 0.794 and independent transfer retained 0.814. Residualisation therefore measured removal of design-aligned variation, not the technical share of the original classifier.

The same simulation provided a positive control. At lower design-label association ($\beta=1, \delta=0.5, \rho=0.25$), mean information fraction was 0.933, design-only AUC was 0.626, and the largest loss under residualisation or restriction was 0.028. External AUC was not lower than internal AUC, and a complete-pipeline permutation test gave $p=0.003984$. All five checks therefore passed in a setting where biological signal was recoverable and portable.

2.3. The GSE135779 design signal survives complete-pipeline null tests

The high complete-design AUC in GSE135779 could, in principle, reflect a flexible pipeline applied to only 44 donors. We therefore repeated the full nested imputation, encoding, scaling, regularisation selection and outer prediction pipeline under 1,000 label permutations per design block. The observed complete design AUC of 0.952 exceeded every permutation except the finite-sample correction required for an empirical upper-tail $p=0.000999$ (Figure 3A). An independent null that shuffled each design column separately while preserving its marginal distribution gave $p=0.003984$ over 250 repetitions. Collection year, sequencing QC and demographics individually produced weaker evidence ($p=0.017$, 0.040 and 0.032 under label permutation); batch alone was null (AUC 0.499, $p=0.470$). The combined label channel was distributed across variables rather than carried by one categorical batch.

Model flexibility did not explain the result. Random-forest and gradient-boosting design-only classifiers achieved mean AUC 0.824 and 0.815, respectively, under the same repeated outer donor splits. Both exceeded chance but remained below the regularised logistic model, indicating that the strongest recorded design-label structure was predominantly low-dimensional rather than an artefact of a flexible nonlinear learner.

The information calculation reached the same conclusion from a different direction. Batch left 87.9% of centred label variation available after adjustment. The complete 21-column design matrix left only 20.3%, corresponding to a relative variance inflation of 4.93 before any molecular feature was considered (Figure 3B). Thirty-six donors remained in mixed-label sequencing batches and 32 in mixed-label collection years, so overlap-based analyses were still possible; adjustment of the complete design nevertheless approached the low-information regime identified by simulation.

2.4. The representations remember the design more accurately than the disease

We next asked whether the donor representations retained the recorded design identity. Under the same fold-contained pipeline, each representation predicted case/control status and every batch one-versus-rest (Figure 4).

In GSE174188, Geneformer, HVG pseudobulk and PCA pseudobulk classified disease at AUC 0.982, 0.977 and 0.975. Their best processing-wave AUCs were 0.9997, 0.999 and 0.997. In GSE135779, disease AUCs were 0.870, 0.933 and 0.945, while the best individual-batch AUCs were 0.993, 0.961 and 0.961. Near-perfect batch recognition was therefore not a peculiarity of one representation family or one cohort.

This result establishes exposure, not attribution. A representation that predicts batch can still contain disease information, and a batch-aware classifier need not use batch as its disease decision rule. The distinction is resolved by restriction, not by correlation alone.

2.5. An internal negative control separates design exposure from shortcut use

GSE135779 supplied a direct counterexample to the common inference that “batch is predictable from the representation” means “the disease classifier uses batch.” The best one-versus-rest batch AUC was 0.993 for Geneformer and 0.961 for both pseudobulk representations. Yet batch alone predicted disease at AUC 0.499, and

fold-contained batch residualisation changed disease AUC by only +0.002, -0.002 and -0.009. Thus representation-to-design predictability documents exposure, but dependence of the disease decision must be demonstrated separately.

2.6. Cell-type composition reproduces the design-adjustment contradiction

We next asked whether the identifiability problem extended beyond high-dimensional molecular representations. For every GSE174188 donor, we aggregated the same CD4-positive cells into naive, effector-memory, regulatory and other-or-unassigned fractions. Raw proportions, centred-log-ratio composition and each representation augmented with log cell yield were evaluated with the locked 20-repeat donor pipeline.

Composition alone was a strong disease classifier. Mean AUC was 0.899 for raw proportions, 0.903 for centred-log-ratio composition and 0.915 after adding cell yield (Figure 3C). Yet each composition vector also predicted processing-wave identity: one-versus-rest wave AUCs reached 0.803-0.872 for wave 1 and 0.817-0.847 for wave 4. Fold-contained processing-wave residualisation reduced all four disease models to AUC 0.689-0.709.

Restriction again gave a different answer. Within dominant wave 4, observed composition AUC was only 0.027-0.062 below size- and label-matched donor subsets; within pure wave 4, the difference was 0.034-0.037 (Figure 3D). Every 2.5th-97.5th percentile interval crossed zero. Thus a simple biological aggregate reproduced the same empirical pattern as Geneformer and expression pseudobulk: design was readily recoverable, residualisation removed a large disease-aligned component, and observed overlap preserved substantially more discrimination than residualisation implied.

2.7. Disease discrimination survives design restriction

We first restricted GSE174188 to dominant wave 4, the only large near-balanced wave (n=89; 46 cases and 43 controls). Because reducing 261 donors to 89 inevitably lowers performance, we compared the observed stratum with 20 random donor subsets having the same donor count and case/control composition (Figure 5A). Relative to those matched subsets, AUC changed by -0.014 for Geneformer, -0.028 for HVG pseudobulk and -0.049 for PCA pseudobulk.

The stricter analysis retained only donors with at least 99% of analysed cells in wave 4 (n=66; 25 cases and 41 controls). Relative to size- and label-matched random subsets, AUC changed by -0.026, -0.044 and -0.044 (Figure 5B). These are detectable losses, but they are far smaller than the raw full-cohort-to-stratum difference. Most of that raw difference is explained by using fewer donors.

GSE135779 provided an independent design. Removing the all-case B1 batch left 36 donors from batches B2-B6, with both labels represented in every retained batch. Geneformer AUC was 0.846 versus 0.858 in matched random subsets (difference -0.012). Both pseudobulk representations were 0.919 versus 0.916 (difference +0.003; Figure 5C). Thus the disease signal persisted after the most obvious label-exclusive batch was removed.

Across both cohorts, design restriction weakens the claim that internal AUC is a pure technical shortcut. The representations carry design information, but the measured disease separation does not disappear when the clearest design channels are held fixed.

2.8. Residualisation can erase the signal that restriction preserves

Residualisation produced a sharply different conclusion. All unadjusted and adjusted models were recomputed on identical outer folds with the same training-only HVG selection, scaling, PCA construction, regularisation grid and classifier.

In GSE174188, ridge residualisation on processing-wave fractions reduced Geneformer AUC from 0.982 to 0.714, HVG pseudobulk from 0.977 to 0.747 and PCA pseudobulk from 0.975 to 0.735 (losses 0.268, 0.230 and 0.241; Figure 6A). The same design, tested by dominant- or pure-wave restriction with matched donor controls, cost only 0.014-0.049. Residualisation therefore removed five to nineteen times more AUC than the matched restriction discrepancy, depending on representation and stratum. The paired split-bootstrap difference between residualisation loss and matched restriction loss was 0.186-0.255 across the six comparisons; every descriptive

2.5th-97.5th percentile interval remained above zero (Figure 6F).

GSE135779 revealed why. Batch alone had a cross-validated disease AUC of 0.499, and batch residualisation changed representation AUC by +0.002, -0.002 and -0.009 (Figure 6B). Collection year had AUC 0.721 and its removal cost 0.005-0.039. The complete measured design block had AUC 0.952; projecting it out collapsed Geneformer, HVG pseudobulk and PCA pseudobulk to 0.516, 0.526 and 0.515, removing 0.354, 0.407 and 0.430 AUC (Figure 6C-D). Yet exclusion of the all-case batch had changed matched performance by at most 0.012.

The conclusion was not specific to a linear residualiser. A training-fold random forest fitted from the complete design to each representation retained AUC 0.748, 0.787 and 0.795, corresponding to losses of 0.122-0.150 (Figure 6E). In contrast, training-only batch location adjustment retained AUC 0.864-0.911, and full-design overlap weighting retained 0.859-0.937. The method matters quantitatively, but every full-design feature-removal arm attenuated discrimination much more than the batch-only negative control.

The projected covariate block predicts how much discrimination residualisation will erase. This is expected algebraically: when design predicts the label, the component of a representation predictable from design includes disease-aligned variation. Residualisation cannot determine whether that variation is technical, biological or both. It answers how much discrimination survives deleting design-aligned signal, not how much of the original discrimination was caused by technical batch.

This empirical discrepancy is the paper's central finding. A large post-residualisation loss is not evidence that the original AUC was spurious. Without overlap-stratum evaluation, it is equally compatible with successful removal of a shortcut and destructive removal of outcome-aligned biology.

In a separate synthetic sensitivity analysis, linear confound regression followed by a random-forest classifier did not inflate null performance in our Gaussian construction: fold-contained residualisation gave mean AUC 0.500 and global residualisation 0.438. This does not rule out confound leakage under skewed or limited-precision features [28]; it shows that the empirical attenuation reported here was not accompanied by that failure mode in the tested construction.

2.9. Transfer degrades across processing waves more than across cohorts

Design restriction preserved strong within-cohort discrimination, but transfer weakened it at every level tested (Figure 7). Training on processing waves 2+3 and testing on wave 4 in GSE174188 produced AUC 0.842, 0.811 and 0.806 for Geneformer, HVG pseudobulk and PCA pseudobulk. Reversing the direction produced 0.919, 0.926 and 0.821. Wave 1 could not be evaluated as a target because it contained no cases.

Strict source-only transfer from the 26-donor GSE285773 cohort to the independent 261-donor GSE174188 target yielded AUC 0.884 [0.843-0.920] for Geneformer, 0.919 [0.884-0.946] for HVG pseudobulk and 0.926 [0.895-0.953] for PCA pseudobulk. On the same 261 target donors, both pseudobulk methods exceeded Geneformer after paired DeLong testing and Benjamini-Hochberg adjustment ($q=0.0151$ and $q=0.000603$). The reverse direction retained the same ranking but was underpowered at 26 target donors.

Restricting the GSE174188 source to dominant wave 4 did not improve transfer to GSE285773. Target AUC fell to 0.794, 0.894 and 0.888 and was no better than size- and label-matched random source subsets. More varied source donors transferred better than the apparently cleaner restricted source.

The cross-cohort result has a direct translational interpretation without implying clinical readiness: a molecular classifier is useful beyond its development cohort only if patient ranking survives changes in age distribution, recruitment and technical context. Internal AUC did not predict that property here.

2.10. Learned pooling does not repair the transfer gap

Mean pooling might suppress informative cell heterogeneity, so we tested DeepSets, gated-attention multiple-instance learning and pooling by multi-head attention [20-22]. All operators used the same frozen cell embeddings, cap-500 cells per donor, a source-fitted whitened PCA32 projection and source-only hyperparameter selection. Independent target data were never used to select a model (Figure 8).

In the 261-donor target, none of the learned operators beat ordinary mean pooling. Relative to the 1,152-dimensional Geneformer donor mean, their paired AUC differences were -0.133, -0.055 and -0.042. Relative to a donor mean in the identical PCA32 space, the differences were -0.077, +0.001 and +0.014; none of the positive differences was significant. All learned operators remained below both pseudobulk representations. In the 26-donor target, no learned-pooling comparison was significant.

This is a useful negative method result. The external gap was not repaired by adding permutation-invariant capacity to the frozen embedding. A more expressive pooling function can fit a source cohort without creating a transportable patient representation.

2.11. An executable validity standard

The results support five minimum requirements for patient-level single-cell classification:

- **Quantify design-label estimability.** Report design-only disease AUC, $1-R^2(Y \sim D)$, design rank and overlap counts before interpreting molecular discrimination.
- **Test the complete pipeline under null data.** Label permutation must repeat imputation, encoding, feature handling, hyperparameter selection and outer prediction rather than test only a final fixed score.
- **Report representation-to-design predictability.** A representation that retains design identity at near-perfect AUC has documented exposure to cohort structure, even if attribution remains unresolved.
- **Pair residualisation with design restriction and matched donor controls.** Residualisation alone is uninterpretable under label-design collinearity.
- **Require a source-only external target.** Target labels must not choose feature transformations, and internal performance must be reported beside external performance.

These checks are inexpensive relative to embedding generation or deep pooling. They convert “the model achieved high AUC” into five separable statements: the design-label contrast is estimable, the pipeline exceeds its null, the design is visible, the signal survives observed overlap, and the ranking transfers.

3. Discussion

This study identifies an estimability boundary that representation leaderboards do not measure. When study design predicts the disease label, the biological contrast available after design adjustment shrinks as $1-R^2(Y \sim D)$; at perfect collinearity, biological and technical explanations are observationally equivalent. No increase in model capacity can recover information absent from the acquisition design. The simulations establish this result across controlled mechanisms, while the two SLE cohorts show that commonly reused patient-level benchmarks can occupy the low-information regime.

The decisive empirical comparison is between two adjustment strategies. Restriction asks whether discrimination survives among donors observed under the same or overlapping design. Residualisation deletes coordinates predictable from a chosen design block. When design and disease are collinear, these operations target different quantities. The two-cohort result makes the distinction visible: mixed-batch restriction preserved GSE135779 performance, while residualising the complete design block reduced every representation to chance. In GSE174188, wave restriction caused modest matched losses, while wave residualisation removed roughly one quarter of the AUC scale. Random-forest residualisation attenuated less than ridge but preserved the same ordering, whereas overlap weighting and batch location adjustment retained substantially more discrimination. CD4-subtype composition independently reproduced the discrepancy, showing that the problem is not peculiar to foundation-model embeddings.

The result is more consequential than a contest between Geneformer and pseudobulk. Recent benchmarks already show that simple expression baselines can outperform frozen foundation-model embeddings [13,14], and patient-level representation learning is an active field [8-11]. Our contribution is to formalise and measure why an internal comparison can remain scientifically ambiguous even when donor leakage is prevented, folds are strict and adjustment is performed inside each fold. The design-only permutation test confirms that the empirical signal is not an artefact of a small-sample modelling pipeline; the information fraction measures how little outcome contrast remains after adjustment. Correct computation does not repair a non-identifiable design.

The external results identify what can be concluded. Strong discrimination persisted inside design-restricted strata, supporting a reproducible disease-aligned signal. The same representations lost performance across processing waves and cohorts, showing that the signal was not sufficiently invariant. Pseudobulk converted additional source donors into transferable information more effectively than frozen Geneformer in the evaluated CD4 setting, and learned pooling did not reverse the ordering. These are not contradictory findings: internal separability can be real while external generalisation remains inadequate.

For clinical translation, this reframes representation selection as a cross-centre validity problem. A classifier that ranks patients inside one retrospective cohort but changes under a new acquisition design is not ready for probability interpretation or individual decisions. The present external AUCs quantify ranking, not calibrated risk. The immediate deliverable is therefore not a diagnostic model but a validation protocol that future diagnostic models should pass.

4. Limitations

This study is confined to three public SLE cohorts, frozen Geneformer final-layer CLS embeddings, fixed expression baselines and a small set of learned pooling operators. The estimability scale is derived for an additive model and does not claim that AUC must change monotonically under every data-generating process. The nonlinear sensitivity arm used one tree-ensemble residualiser and two design-only classifiers; it is not an exhaustive comparison of adjustment algorithms. The simulation isolates one binary design variable, whereas real cohorts can contain interacting, continuous and unrecorded design channels. The CD4 composition extension uses the cell labels distributed with GSE174188 and does not re-annotate subtypes independently. The overlap strata are smaller than the full cohorts, and matched random subsets separate sample-size loss from design restriction without identifying a causal technical effect. Recorded batch can combine recruitment source, age, ancestry, disease severity and laboratory processing; our result is non-attributability, not proof that the retained signal is purely biological. GSE285773 lacks recoverable donor processing-batch metadata. External bootstrap intervals condition on one fitted source model, the 26-donor target has limited power, and external probabilities are not clinically calibrated. Geneformer pretraining overlap with individual benchmark cells was not tested. These boundaries do not weaken the central empirical finding: residualisation, restriction and transfer produced different answers on the same donors and cannot substitute for one another.

5. Methods

5.1. Cohorts, outcomes and statistical unit

We analysed GSE135779 (44 childhood donors; 33 SLE, 11 controls), the CD4-positive alpha-beta T-cell subset of GSE174188 (261 donors; 162 SLE, 99 controls) and GSE285773 CD4-positive T cells (26 donors; 16 SLE, 10 controls) [1-3]. The donor was the independent unit in every supervised analysis. Cells from one donor never crossed training and test partitions.

5.2. Recovery of GSE135779 design metadata

GSE135779 batch, collection year, age, sex, race, ethnicity and clinical variables were read from the original Supplementary Table 1b; per-library sequencing metrics were read from Supplementary Table 1c [1]. Study names were linked to analysis donor identifiers through the GEO title/accession map. All 44 donors mapped uniquely. The restoration script writes the donor table, batch-by-label, year-by-label and batch-by-year cross-tabs,

and SHA256 hashes of every source file.

For GSE174188, per-cell `Processing_Cohort` was read from the distributed CELLxGENE object and reduced to donor fractions over four waves. Dominant wave was the maximum-fraction wave; a pure-wave donor had at least 99% of analysed cells in one wave.

5.3. Design-label information and observational equivalence

For centred binary label vector Y , intercept-only projection M_1 , recorded design matrix D including an intercept and residual projection M_D , we defined the design-adjusted label-information fraction as $I_D = (Y^T M_D Y) / (Y^T M_1 Y) = 1 - R^2(Y \sim D)$. Values below floating-point tolerance were set to zero. Relative variance inflation was $1/I_D$, reported as infinite at zero information. Design rank and numbers of mixed-label levels and donors were reported beside this measure.

Observational equivalence was demonstrated constructively. At $D=Y$, the models $X = \beta Y + \epsilon$ and $X = \delta D + \epsilon$ generate identical observations when $\beta = \delta$, despite assigning the entire effect to different mechanisms. The numerical test used identical Gaussian noise and required maximum absolute difference zero.

5.4. Identifiability simulation

The additive donor simulation generated 240 source and 2,000 independent target donors with 20 features. The first coordinate received biological effect $\beta \in \{0, 0.5, 1, 2\}$ and technical effect $\delta \in \{0, 0.5, 1, 2\}$. A binary design variable was generated with association $\rho \in \{0, 0.25, 0.5, 0.75, 0.9, 0.98, 1\}$ to the balanced disease label; target design was independent of disease. Each of 112 parameter cells used 100 deterministic replicates.

Balanced logistic regression compared unadjusted source cross-validation, training-fold residualisation, maximum-overlap design restriction and source-only target prediction. Restriction was evaluated in one design stratum selected for maximum label overlap; probabilities from separately fitted strata were not pooled. Biological-coefficient recovery used ordinary least squares on disease and design and was undefined at rank deficiency. All 11,200 replicate rows and the exact observational-equivalence check were retained.

The prespecified low-entanglement positive control used $\beta = 1, \delta = 0.5, \rho = 0.25$. Five illustrative pass criteria were evaluated: information fraction at least 0.80; complete-pipeline permutation $p \leq 0.01$; design-only AUC at most 0.70; maximum residualisation or restriction loss at most 0.05; and internal-minus-external AUC at most 0.05. The thresholds define an interpretable demonstration of the protocol rather than a universal acceptance rule.

5.5. Complete-pipeline design nulls

For every GSE135779 design block, 1,000 label permutations repeated the entire nested donor pipeline, including fold construction, imputation, one-hot encoding, scaling, inner regularisation selection and outer prediction. A second null used 250 independent shuffles of every design column while keeping disease labels fixed and preserving each column's marginal distribution. Empirical upper-tail probabilities used $(1 + \sum I[T_b \geq T_{\{obs\}}]) / (B + 1)$. Null draws used eight parallel workers; no molecular feature entered these tests.

5.6. CD4-subtype composition

The GSE174188 CELLxGENE object was restricted to the same CD4-positive alpha-beta T-cell population and 261 donors used in the molecular analysis. Per-donor counts were formed for `T4_naive`, `T4_em`, `T4_reg` and all other or unassigned labels. We evaluated raw closed proportions and centred-log-ratio coordinates. For CLR, 0.5 was added to each donor-category count before closure; donor mean log abundance was subtracted and the final coordinate omitted. Each representation was evaluated with and without log total CD4-cell yield.

The four composition variants used the locked repeated donor classifier. Processing wave was predicted one-versus-rest. Fold-contained ridge residualisation used processing-wave fractions. Dominant-wave-4 and at-least-99%-pure-wave-4 restrictions were compared with donor subsets matched exactly for size and case/control count using the same 20 seeds as the main analysis.

5.7. Frozen Geneformer embeddings

Cell embeddings used Geneformer V2-316M at repository revision

04c2b2e84da7c0f385c3f9ad8f3ec24bab6650e5 [4]. The archived checkpoint, configuration, token dictionary and gene-median dictionary SHA256 hashes are provided in the methods manifest. Gene identifiers were version-trimmed and mapped to the V2 vocabulary. Within each cell, the ranking value was $\text{raw count} / \text{total count} * 10,000 / \text{Genecorpus-104M gene median}$; genes were sorted descending. Sequences used V2 start/end token IDs 2/3, padding ID 0 and maximum length 4096, leaving at most 4094 ranked genes.

The model output was the final `last_hidden_state`; token position 0 was retained as the 1,152-dimensional cell CLS vector. The encoder was frozen. Donor representations were coordinate-wise means of sampled cell vectors. Sampling without replacement used integer seed 1 and caps of 500 cells per donor for GSE135779 and 1,000 for GSE174188 and GSE285773. The embedding environment used Python 3.10, PyTorch 2.5.1 and Transformers 4.46.3.

5.8. Pseudobulk representations

Stored donor-by-gene values were $\log_{1p}(\text{CPM})$. Within every outer training fold, gene variance was computed on training donors, the 4,000 highest-variance genes were selected, and coordinates were standardised by training-donor mean and standard deviation. HVG pseudobulk used these values directly. PCA pseudobulk fitted up to 30 components on the training values and applied the fitted map to held-out donors.

For cross-cohort transfer, trimmed gene identifiers were intersected before source-only HVG selection. Feature means, variances, selected genes, scaler, PCA, classifier and regularisation were fitted on source donors and applied unchanged to the target.

5.9. Repeated donor-level evaluation

Internal analyses used 20 repeated stratified five-fold donor splits with integer seeds 20260801-20260820. The classifier was balanced logistic regression with the liblinear solver and `max_iter=20,000`. Inverse regularisation strength was selected from 10^{-4} through 10^4 in decade steps using five-fold resampling within each outer training set. The full outer-training representation was used for this inner regularisation selection; no outer test donor entered HVG selection, scaling, PCA or classifier fitting.

Out-of-fold donor predictions were pooled within each repeat. We report mean AUC and the 2.5th-97.5th percentile of repeat-level AUCs. This is a descriptive split-sensitivity range, not a population confidence interval or a test based on treating correlated folds or repeats as independent [31].

5.10. Design-only and representation-to-design prediction

Numeric design variables were median-imputed and standardised inside the outer training fold. Categorical variables were most-frequent-imputed and one-hot encoded, with held-out unknown levels ignored. The same logistic model and regularisation grid were used. GSE135779 blocks were batch, collection year, sequencing QC, demographics and their union. GSE174188 blocks were processing-wave fractions, sequencing QC, demographics and their union.

Each molecular representation also predicted every observed batch one-versus-rest under five repeated donor-level cross-validations. The number of folds was capped by the minority batch count.

As a model-flexibility sensitivity analysis, the complete GSE135779 design block was also evaluated with random forests (256 trees, maximum depth 4, minimum leaf size 3) and histogram gradient boosting (150 iterations, learning rate 0.05, seven terminal leaves, minimum leaf size 5). Encoding was fitted inside every outer training fold; these arms used fixed hyperparameters and were not selected against held-out donors.

5.11. Fold-contained residualisation

Unadjusted and residualised models shared outer folds, feature construction, scaling, PCA and classifier selection. Within each outer training fold, design variables were encoded as above and a multivariate ridge regression ($\alpha=1$, intercept included) mapped design to every raw representation coordinate. Training and held-out coordinates were replaced by observed minus predicted values, after which the identical representation-specific pipeline was fitted to training residuals and applied to held-out residuals. The disease label was not included in the residualiser.

5.12. Nonlinear adjustment, weighting and attenuation uncertainty

The GSE135779 complete-design sensitivity was repeated with a multivariate random-forest residualiser fitted only on outer-training donors (96 trees, maximum depth 4, minimum leaf size 3). Pseudobulk genes were selected by training-donor variance before this high-dimensional fit. A separate batch-location arm subtracted training-batch mean offsets from training and held-out features; because it did not apply empirical-Bayes scale shrinkage, we interpret it as ComBat-style location adjustment rather than ComBat. A third arm fitted a training-only logistic disease propensity model from the complete design, clipped propensities to 0.025-0.975 and trained the representation classifier with overlap weights.

For the central GSE174188 comparison, residualisation loss was paired by split seed. We bootstrapped the 20 paired repeat-level losses 5,000 times and subtracted the locked matched-restriction discrepancy for dominant or pure wave 4. These intervals quantify split sensitivity conditional on the analysed donors and matched-control estimate; they are not population confidence intervals.

To probe confound leakage with a nonlinear downstream learner [28], 100 Gaussian simulations used no biological effect, a technical effect of one and design-label association 0.75. Random-forest disease prediction was compared on raw features, features residualised globally and features residualised inside each training fold.

5.13. Design restriction and matched controls

GSE174188 was restricted to dominant wave 4 and separately to pure wave 4. GSE135779 excluded all-case batch B1 and retained B2-B6, each of which contained both labels. The complete internal pipeline was rerun after restriction.

For every observed stratum, 20 donor subsets were drawn without replacement from the full cohort with the identical donor count and case/control count. Each draw used one prespecified split seed. The observed-minus-matched difference quantifies whether the stratum is harder than expected from reduced donor number and label composition.

5.14. Cross-batch and cross-cohort transfer

Cross-batch analyses trained on waves 2+3 and evaluated wave 4, then reversed the direction. Cross-cohort transfer fitted every statistic on the source cohort. Target labels were accessed only after prediction.

Target AUC intervals used 5,000 case/control-stratified donor bootstrap resamples and the percentile method. Paired AUC comparisons used DeLong tests on the identical target donors [23] and Benjamini-Hochberg correction within declared method families [24].

5.15. Learned pooling

DeepSets, gated-attention multiple-instance learning and pooling by multi-head attention operated on cap-500 frozen cell embeddings [20-22]. Cell standardisation, a whitened source PCA32 projection, network weights and hyperparameters were source-fitted. Hidden widths were 16 or 32, weight decay 0.01 or 0.1, and Adam used a learning rate of 0.02 for 150 epochs. Five-fold source-donor cross-validation selected the configuration; three initialisations were fitted on all source donors and target probabilities were averaged. An ordinary donor mean in the identical PCA32 space isolated the contribution of learned pooling.

The PCA32 projection used for hyperparameter selection was fitted once on all source cells before source-donor folds. It never used target cells or labels. Consequently, source-internal pooling AUC was not used as an unbiased

generalisation estimate; only independent-target and paired target-donor comparisons support the learned-pooling conclusion.

5.16. Software and provenance

The locked donor-level reanalysis used Python 3.13.7, NumPy 2.4.6, pandas 3.0.3, SciPy 1.18.0, scikit-learn 1.9.0, PyArrow 21.0.0 and joblib 1.5.3. Input hashes, seeds, parameter grids, donor-level predictions, complete null distributions and output manifests accompany the analysis. The identifiability extension contains 18 automated validation checks covering simulation cardinality, exact observational equivalence, information monotonicity, empirical donor counts, composition attenuation and publication-file completeness.

6. Data and code availability

The source datasets are publicly available through GEO accessions GSE135779, GSE174188 and GSE285773. The MIT-licensed public repository is available at <https://github.com/LightChainr/rheumlens>, and its permanent Zenodo record is <https://doi.org/10.5281/zenodo.20813922>. The complete versioned research object submitted for review accompanies this manuscript. It contains restored GSE135779 metadata, donor-level inputs and predictions, analysis scripts, two locked software environments, 11,200 simulation replicates, complete null distributions, robustness outputs, figure source tables, REPRODUCE .md, SHA256SUMS, RELEASE_MANIFEST .json and an executable package validator. Public raw single-cell matrices remain at their GEO accessions.

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Figure 1 Outcome labels are entangled with study design in two widely reused SLE cohorts

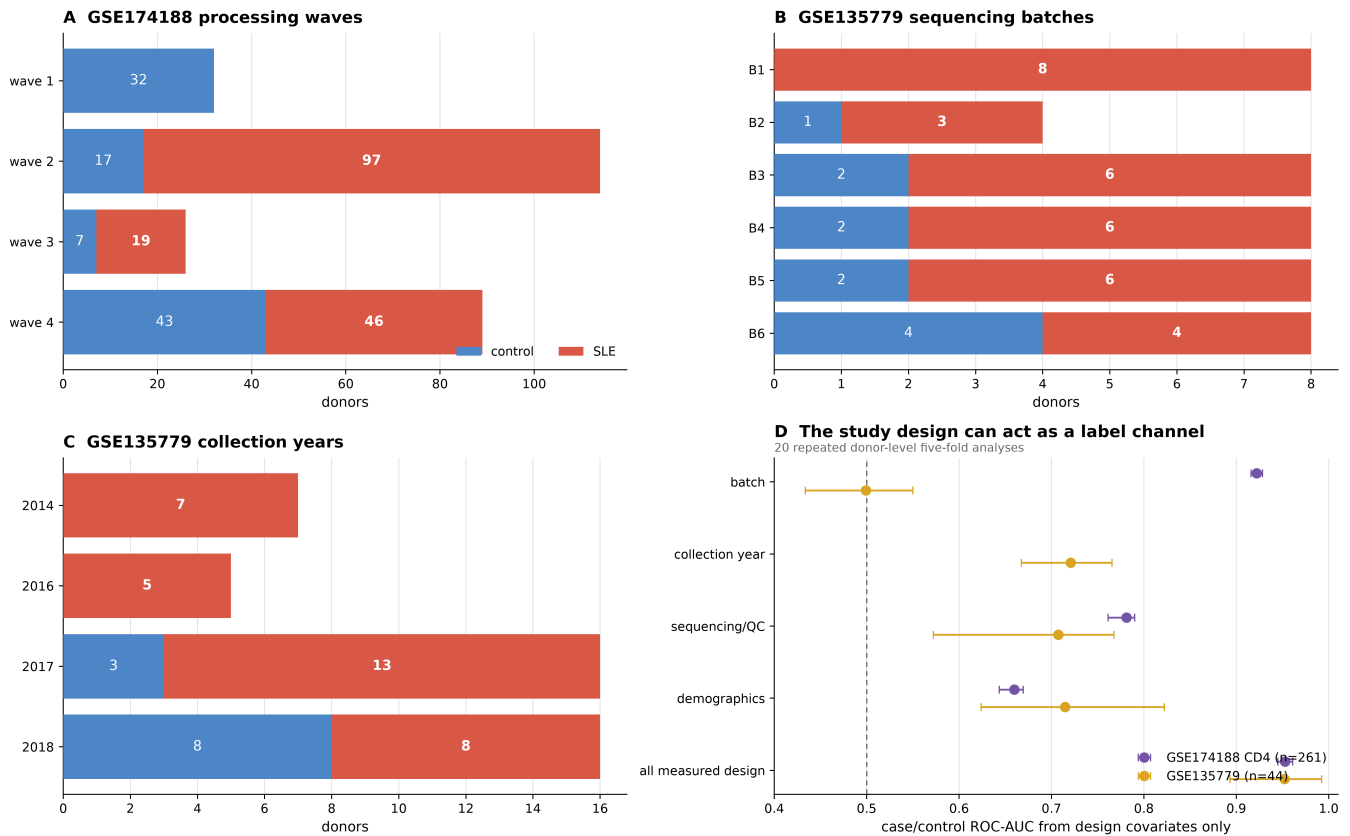
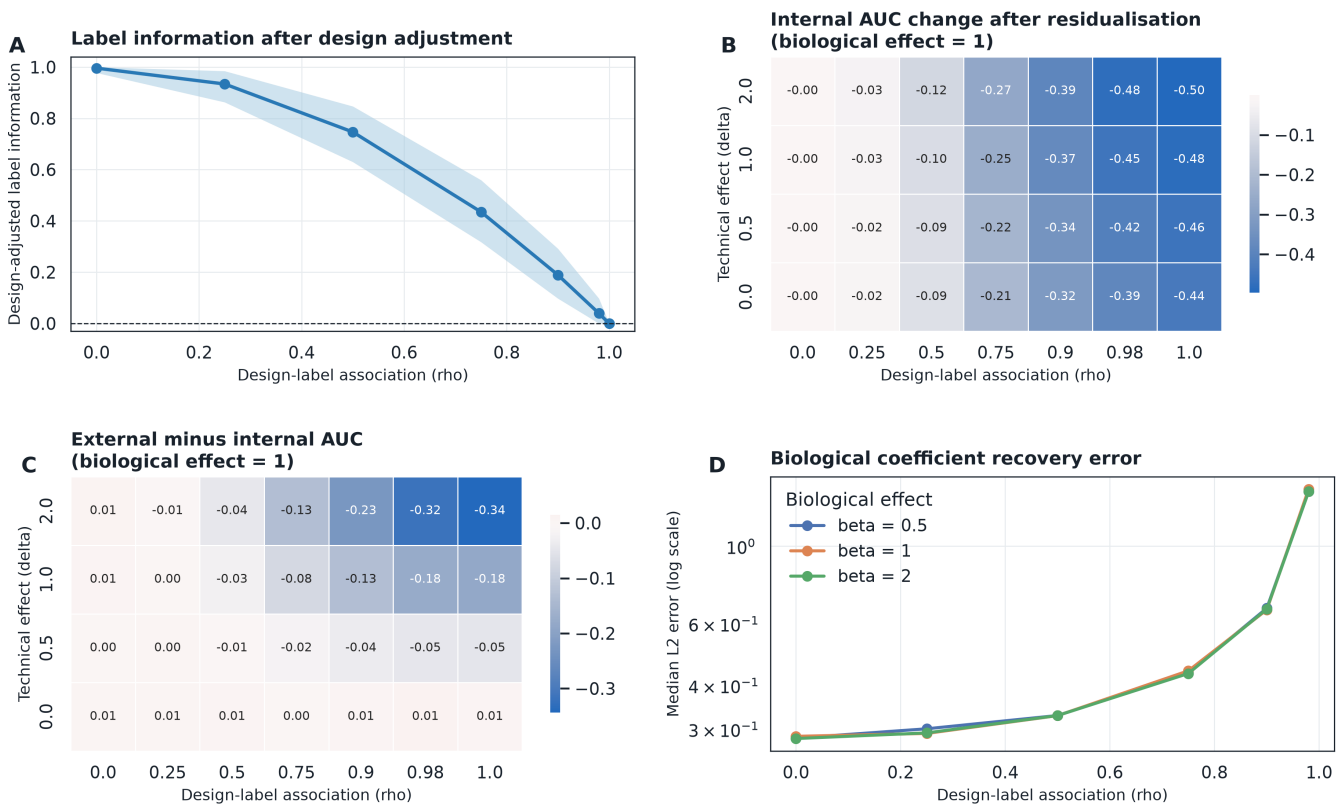


Figure 1. Outcome labels are entangled with study design in two widely reused SLE cohorts (A) Case and control counts across the four processing waves represented among the analysed GSE174188 CD4-positive alpha-beta T-cell donors. Wave 1 contains no SLE cases, whereas wave 2 is dominated by cases. (B) Case and control counts across the six restored GSE135779 sequencing batches. Batch B1 contains only cases; batches B2-B6 contain both labels. (C) Case and control counts by GSE135779 collection year. (D) Repeated donor-level cross-validated ROC-AUC obtained using recorded design variables alone. Points denote mean AUC across 20 repeated stratified five-fold analyses and bars denote the 2.5th-97.5th percentile of repeat-level AUCs. The vertical dashed line marks chance performance.

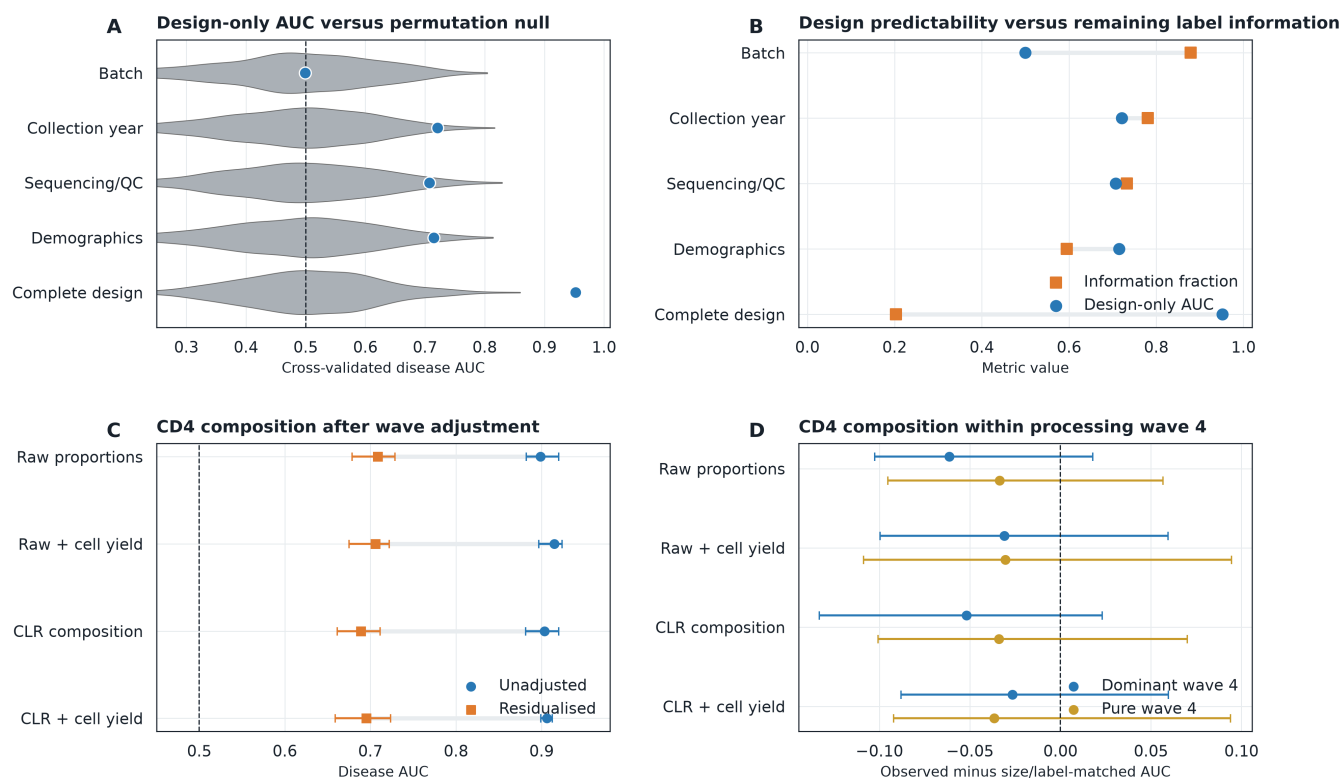
Simulated design-label identifiability landscape



Donor-level additive simulation; 100 replicates per parameter cell. External targets set $\rho=0$. Heatmaps report mean AUC differences.

Figure 2. Design-label association determines how much disease contrast remains estimable (A) Mean design-adjusted label-information fraction and 2.5th-97.5th percentile across 100 replicates at each level of design-label association. (B) Change in internal AUC after fold-contained residualisation when the biological effect equals one, across technical-effect magnitudes and design-label associations. (C) Independent-target minus internal AUC under the same parameter grid; target design was independent of disease. (D) Median error in recovery of the biological coefficient. The simulation crossed four biological effects, four technical effects and seven design-label associations for 11,200 replicates. At perfect collinearity, biological-only and technical-only mechanisms are observationally equivalent.

Empirical identifiability diagnostics and CD4 composition extension



GSE135779: 44 donors and full-pipeline nulls. GSE174188: 261 donors; intervals are 2.5th-97.5th percentiles across 20 split seeds.

Figure 3. Empirical design diagnostics reproduce the simulated identifiability failure (A) Observed GSE135779 design-only AUCs over full-pipeline label-permutation distributions. The complete design reached AUC 0.952 with empirical upper-tail $p=0.000999$. (B) Design-only disease AUC and the fraction of label information remaining after linear design adjustment. (C) GSE174188 disease AUC for raw and centred-log-ratio CD4-subtype composition, with or without log cell yield, before and after fold-contained processing-wave residualisation. Points denote means and bars the 2.5th-97.5th percentile across 20 repeated donor splits. (D) Difference between observed composition AUC within dominant or pure wave 4 and AUC from donor subsets matched for size and case-control count. Intervals crossing zero indicate that the restricted stratum was not consistently harder than its matched controls.

Figure 2 High disease discrimination coexists with near-perfect design recognition

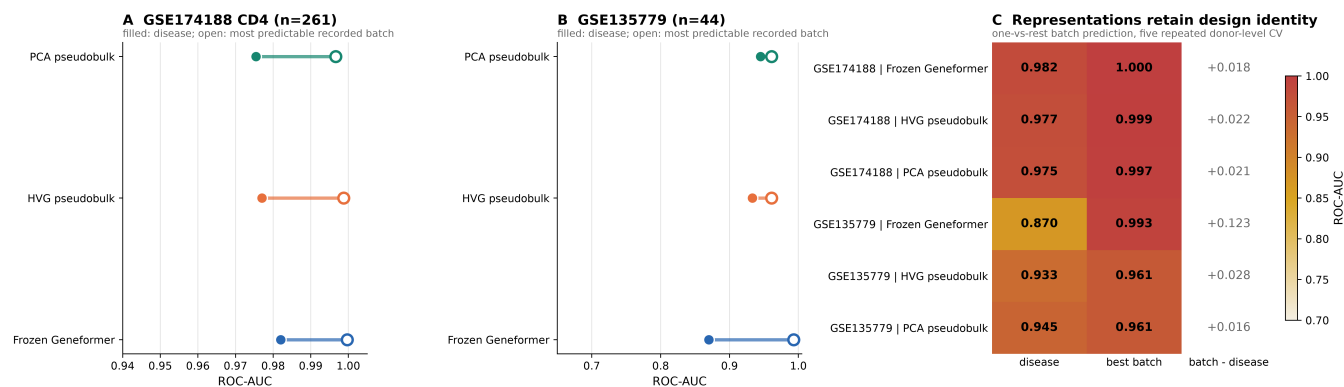


Figure 4. High disease discrimination coexists with near-perfect design recognition (A-B) Mean repeated-cross-validation ROC-AUC for case-control discrimination (filled points) and for the most predictable recorded batch in a one-versus-rest analysis (open points) using frozen Geneformer donor means, HVG pseudobulk and PCA pseudobulk in GSE174188 (A) and GSE135779 (B). Horizontal segments connect the two tasks for each representation. (C) Matrix of disease AUC, best-batch AUC and their difference. Batch predictability documents design information retained by each representation but does not by itself identify that information as technical or biological.

Figure 3 Disease discrimination survives design restriction after controlling for lost sample size

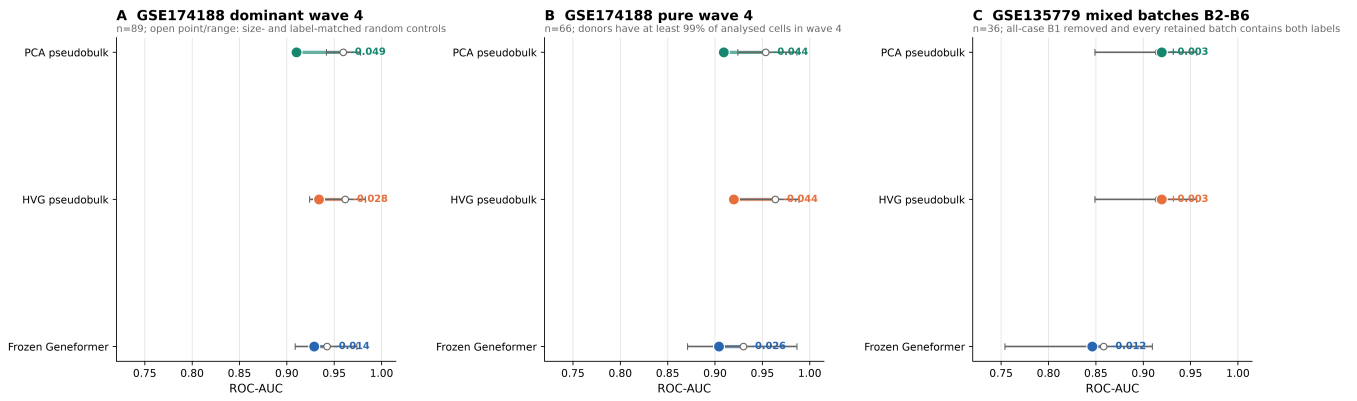


Figure 5. Disease discrimination survives design restriction after controlling for lost sample size (A) GSE174188 donors assigned to dominant processing wave 4 (n=89). (B) The stricter GSE174188 pure-wave-4 subset, requiring at least 99% of analysed cells in wave 4 (n=66). (C) GSE135779 donors from mixed-label batches B2-B6 after excluding the all-case B1 batch (n=36). Coloured points show mean repeated-cross-validation AUC in the observed stratum. Open points and grey bars show the mean and 2.5th-97.5th percentile from 20 random donor subsets matched to donor count and case-control count. Labels report observed minus matched mean AUC.

Figure 6 Residualisation measures sensitivity, not a causal batch effect

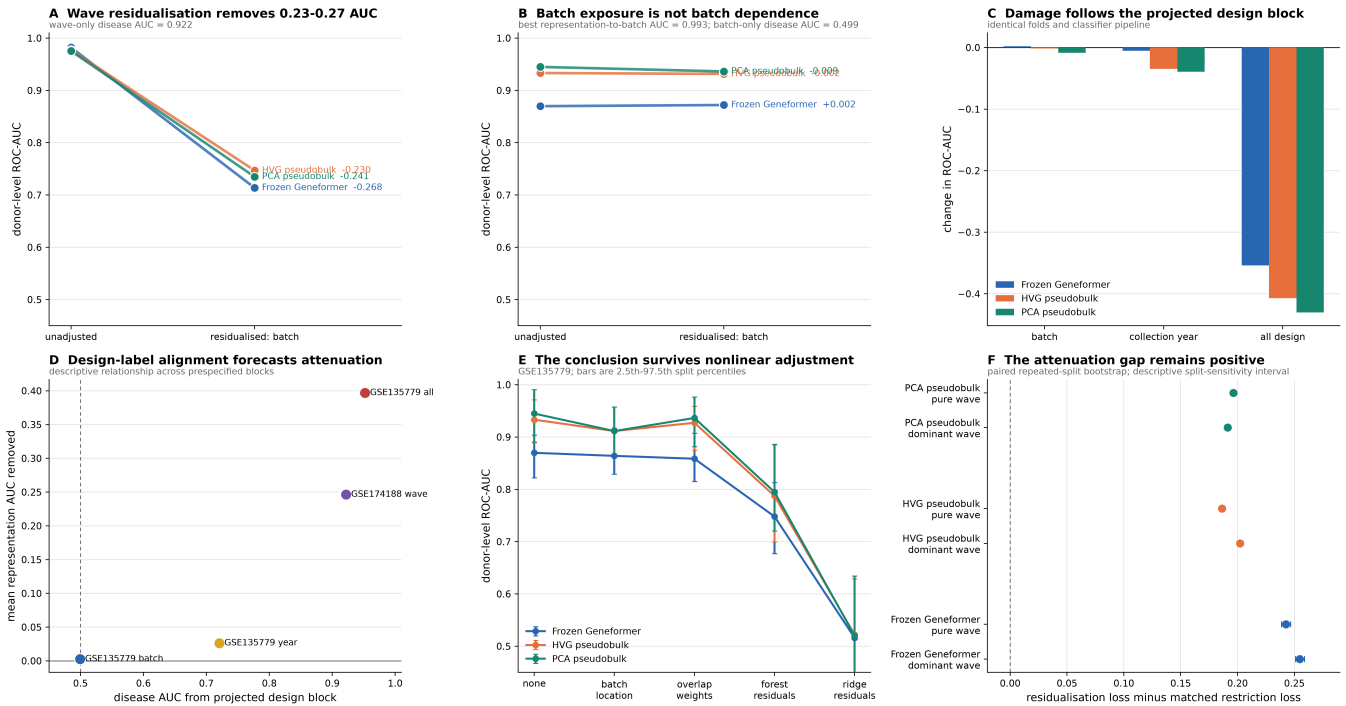


Figure 6. Residualisation measures sensitivity, not a causal batch effect (A) GSE174188 AUC before and after fold-contained residualisation on processing-wave fractions; the projected design block itself predicts disease at AUC 0.922. (B) GSE135779 internal negative control: representations predict batch at AUC up to 0.993, but batch alone is uninformative for disease at AUC 0.499 and batch residualisation changes disease AUC by at most 0.009. (C) Change in GSE135779 AUC after projecting out batch, collection year or the complete measured design block. (D) Across prespecified design blocks, the mean AUC removed from molecular representations increases with disease predictability of the projected block. (E) GSE135779 AUC after training-only batch location adjustment, complete-design overlap weighting, random-forest residualisation or ridge residualisation. Points denote means and bars denote 2.5th-97.5th percentiles across 20 repeated donor splits. (F) Residualisation loss minus the matched restriction loss for dominant and pure wave 4. Bars are descriptive 2.5th-97.5th percentile intervals from 5,000 paired resamples of repeat-level residualisation losses, conditional on the locked matched-restriction estimate. Adjusted and unadjusted estimates use identical outer donor folds and training-only preprocessing.

Figure 5 Generalisation, not internal separability, is where donor-level models break

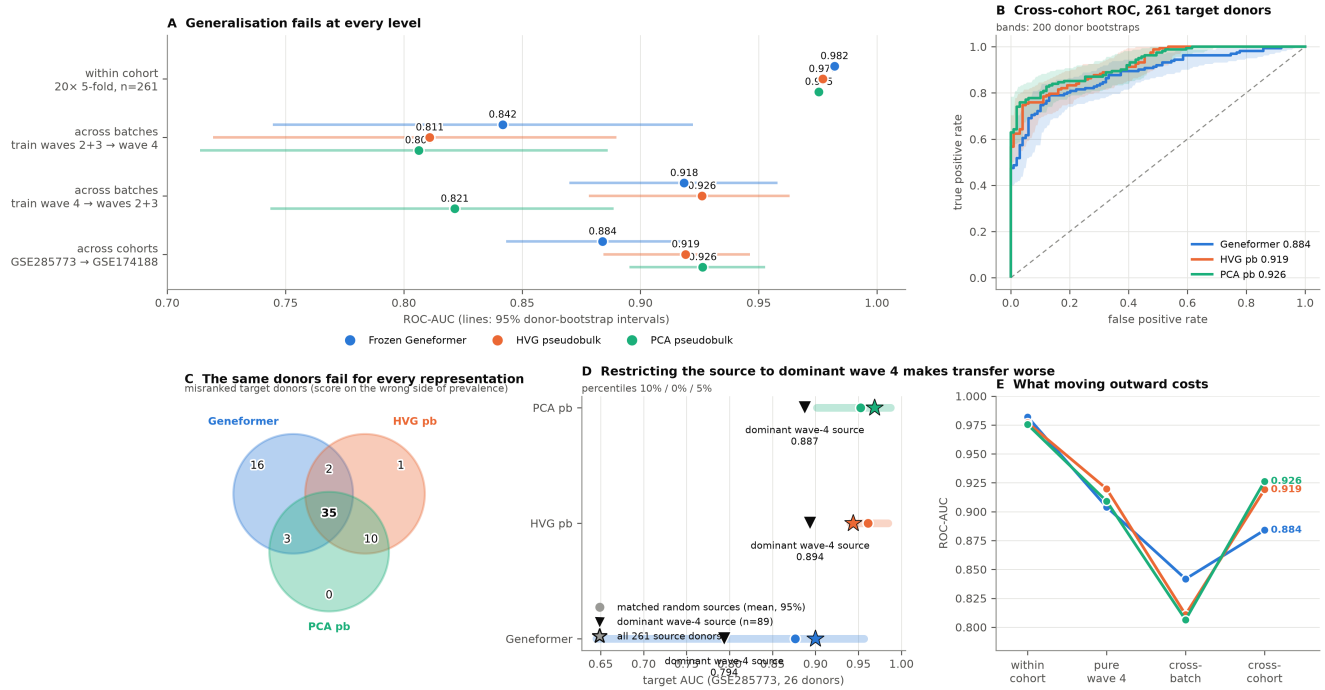


Figure 7. Generalisation, not internal separability, is where donor-level models break (A) Donor-level ROC-AUC for within-cohort evaluation, bidirectional cross-wave evaluation in GSE174188 and strict source-only transfer from GSE285773 to GSE174188. (B) Target ROC curves for the 261-donor GSE174188 cohort after training on GSE285773, with pointwise bands from 200 stratified donor bootstrap resamples. (C) Overlap among target donors whose scores fell on the wrong side of target prevalence for the three representations. (D) Transfer to GSE285773 after restricting the GSE174188 source to dominant wave 4, compared with matched random source subsets and all 261 source donors. (E) AUC across the sequence from within-cohort evaluation to pure-wave, cross-batch and cross-cohort evaluation. Every transformation and classifier in cross-cohort analyses was fitted on source donors only.

Figure 6 Learned pooling does not rescue the frozen representation

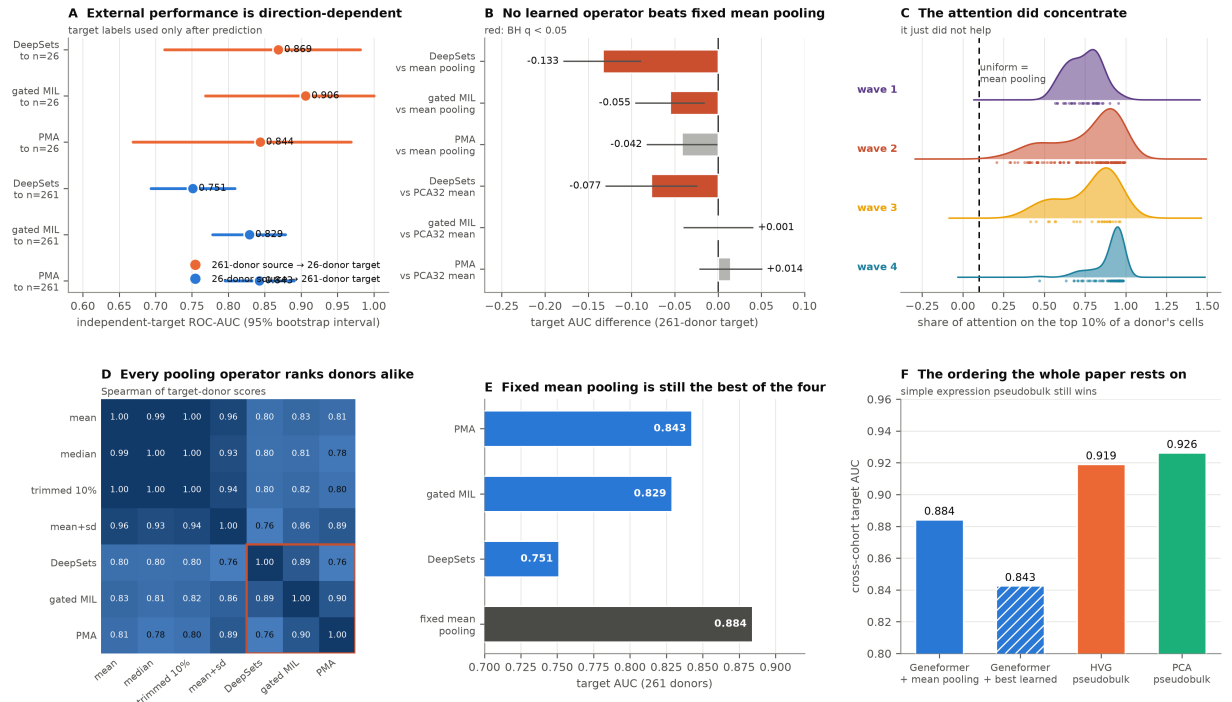


Figure 8. Learned pooling does not rescue the frozen representation (A) Independent-target AUC and 95% stratified donor-bootstrap intervals for DeepSets, gated-attention multiple-instance learning and pooling by multi-head attention in both transfer directions. (B) Paired target-donor AUC differences for the 261-donor target relative to the 1,152-dimensional frozen Geneformer donor mean and a donor mean in the identical source-fitted PCA32 space. Red bars denote Benjamini-Hochberg-adjusted $q < 0.05$. (C) Distribution across source donors of the share of gated-attention mass assigned to the top 10% of cells, stratified by processing wave. (D) Spearman correlations among independent-target donor scores. (E) Target AUC for learned operators and fixed mean pooling in the 261-donor target. (F) Strict source-only cross-cohort AUC for fixed Geneformer mean pooling, the best learned operator and the two expression pseudobulk baselines. Learned pooling was selected using source donors only.